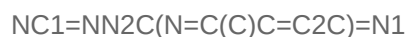
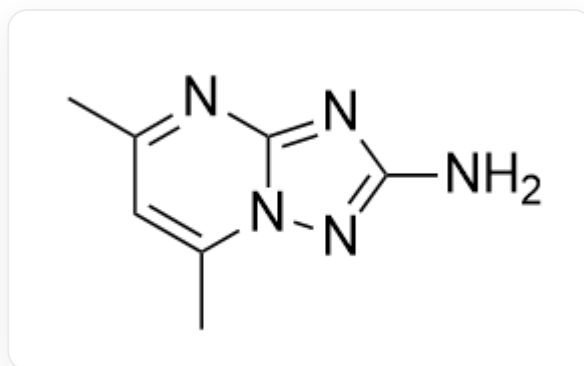


Question

Mix a dilute hydrochloric acid solution of ruthenium trichloride with an ethanol solution of **L** (structure shown below), gently heat at 75°C for 3h , and brick-red powder precipitates;



The powder is recrystallized with an acetone-water mixed solvent, yielding orange-red crystals **A**. The following characterization results are obtained for **A**:

- (1) Magnetic moment measurements show that the magnetic moment of **A** at 20°C is $\mu = 1.9\mu_B$.
- (2) Elemental analysis indicates the mass fractions of elements in **A** are: *C*, 29.52%; *H*, 3.86%; *N*, 24.58%.
- (3) X-ray diffraction reveals that two types of molecules exist in the crystal of **A**: one is a mononuclear octahedral ruthenium complex molecule, which contains 4 hydrogen bonds, with **L** coordinating via the nitrogen atoms linked to two carbon atoms on the five-membered ring; the other is a solvent molecule.
- (4) The ^1H NMR spectrum of **A** in acetone shows that ligand **L** has only one chemical environment.

Which of the following statements about crystal **A** are correct?

1. The electron configuration of *Ru* in **A** in the octahedral field is $t_{2g}^3e_g^2$.
2. The chemical formula of crystal **A** is $\text{RuC}_{14}\text{H}_{22}\text{O}_2\text{N}_{10}\text{Cl}_3$.
3. **A** contains neither acetone nor ethanol molecules.

4. The complex contains $N - H \dots Cl$ hydrogen bonds.

5. The complex contains $N - H \dots O$ hydrogen bonds.

A. 1, 2, 3, 4, 5

B. 1, 2, 3, 4

C. 1, 2, 3, 5

D. 1, 2, 4, 5

E. 1,3,4,5

F. 2, 3, 4, 5

G. 1, 2, 3

H. 1, 2, 4

I. 1, 2, 5

J. 1,3,4

K. 1,3,5

L. 1,4,5

M. 2, 3, 4

N. 2,3,5

O. 2,4,5

P. 3,4,5

Q. None of the above

Answer

Correct Answer: **M**

Detailed Explanation

From the magnetic moment $\mu = \sqrt{n(n+2)} = 1.9\mu_B$, we can obtain the number of unpaired electrons as $n = 1$, so Ru is low spin; the valence electron configuration is $t_{2g}^5e_g^0$, statement 1 is incorrect.

CHECKPOINT

1 PTS

Ru valence electron configuration is $t_{2g}^5e_g^0$, statement 1 is incorrect

The ratio of C, H, N in **A** is: $(29.52/12.01) : (3.86/1.008) : (24.58/14.01) \approx 7 : 11 : 5$. In **L**, $C : N = 7 : 5$. Ethanol and acetone contain carbon but not nitrogen, so **A** does not contain acetone or ethanol. The source of the extra hydrogen should be water;

Therefore, **A** contains Ru, L, H_2O , and may contain Cl and OH ligands.

CHECKPOINT

1 PTS

The ratio of C, H, N in **A** is: $7 : 11 : 5$, from which it is determined that **A** does not contain acetone or ethanol, and contains water molecules

Let the chemical formula of **A** have n ligands **L**, then the molar mass of **A** is:

$$M(A) = 12.01 \text{ g} \cdot \text{mol}^{-1} \times 7n/29.52\% = 284.8n \text{ g} \cdot \text{mol}^{-1}$$

When $n = 1$, the molar mass of 1 ligand **L** and 1 *Ru* has already reached $264.3 \text{ g} \cdot \text{mol}^{-1}$, which is unreasonable;

When $n = 2$, the molar mass $M = 569.6 \text{ g} \cdot \text{mol}^{-1}$; since *Ru* is in the +3 oxidation state, there are 3 *OH* and *Cl* ligands in total; since *Ru* is 6-coordinate and **L** is a monodentate ligand, there should be 1 water molecule within the coordination sphere, so the chemical formula of **A** can be expressed as: $\text{RuL}_2\text{Cl}_{3-x}(\text{OH})_x(\text{H}_2\text{O})_n$

In **L**, $C : H = 7 : 11$, so $x + 2n = 4$; substitute $x = 2, n = 1$ and $x = 0, n = 2$ into the chemical formula in turn, and calculate the molar mass: when $x = 0, n = 2$, the molar mass is $569.8 \text{ g} \cdot \text{mol}^{-1}$, which is consistent with the question, i.e., **A** is $\text{RuL}_2\text{Cl}_3(\text{H}_2\text{O})_2$; then statements 2 and 3 are correct.

CHECKPOINT

3 PTS

A is $\text{RuL}_2\text{Cl}_3(\text{H}_2\text{O})_2$

The actual structure should be $[\text{RuL}_2\text{Cl}_3(\text{H}_2\text{O})] \cdot \text{H}_2\text{O}$. The structure of the complex is

SMILES:

```
Cl[Ru-3]([N+]=C2N(C(C)=CC(C)=N2)N=C1N([H])[H])([O+](H)[H])(Cl)(Cl)
[N+]=C(N=C(C)C=C4C)N4N=C3N([H])[H]
```

According to the description of the NMR test in the question, the amino group in **L** can form $N - H \dots Cl$ hydrogen bonds with chlorine atoms, and the water molecule as a ligand can form $O - H \dots N$ hydrogen bonds with the six-membered ring nitrogen atom in **L**. Therefore, there are $N - H \dots Cl$ and $O - H \dots N$ hydrogen bonds in the complex, statement 4 is correct, and 5 is incorrect.

CHECKPOINT

1 PTS

There are $N - H \dots Cl$ hydrogen bonds, statement 4 is correct

CHECKPOINT

1 PTS

There are $O - H \dots N$ hydrogen bonds, statement 5 is incorrect

Therefore, choose option M.